2. (a)

$$\frac{d}{dt}\int_{a(t)}^{b(t)} f(x)\,dx = \frac{d}{dt}\left(F(b(t)) - F(a(t))\right) = \frac{dF(b(t))}{dt} - \frac{dF(a(t))}{dt}$$

$$= F'(b(t))\,b'(t) - F'(a(t))\,a'(t) = f(b(t))\cdot b'(t) - f(a(t))\cdot a'(t)$$

F is the integral of f.

2. (b) Prove the solution of $u_{tt} - c^2 u_{xx} = 0$ is $u(x,t) = f(x+ct) + g(x-ct)$

(1) the variable $v = u_t + cu_x$.

$$u_{tt} - c^2 u_{xx} = \left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)u = 0$$

Let $v = u_t + cu_x$, we must have $v_t - cv_x = 0$.

This we have two first-order equations $\begin{cases} v_t - cv_x = 0 \\ u_t + cu_x = v \end{cases}$

Since the solution of $au_x + bu_y = 0$ is $u(x,y) = f(bx - ay)$.

From $v_t - cv_x = 0$ we can get $v(x,t) = h(x+ct)$, where h is any function.

So, we only need to solve the equation $u_t + cu_x = h(x+ct)$

Let $w(x,t) := u(x,t) - f(x+ct)$, then $w(x,t)$ satisfies equation

$$w_t + cw_x = u_t + cu_x - cf'(x+ct) - cf'(x+ct) = h(x+ct) - 2cf'(x+ct)$$

Now we choose $f(s) = \frac{h(s)}{2c}$. Hence, $w_t + cw_x = 0$.

Again from the solution of $au_x + bu_y = 0$ is $u(x,y) = f(bx - ay)$.

We can get $w(x,t) = g(x-ct)$, where g is any function.

Notice that h is any function, so f is also any function.

Therefore, the general solution of $u_{tt} - c^2 u_{xx} = 0$ is

$$u(x,t) = f(x+ct) + g(x-ct)$$

(2) the coordinate method

Let $\begin{cases} \xi = x+ct \\ \eta = x-ct \end{cases}$ By the chain rule, we have $\begin{cases} \partial_x = \partial_\xi + \partial_\eta \\ \partial_t = c\partial_\xi - c\partial_\eta \end{cases}$

Therefore, $\partial_t - c\partial_x = -2c\partial_\eta$, $\partial_t + c\partial_x = 2c\partial_\xi$

So, $(\partial_t - c\partial_x)(\partial_t + c\partial_x)u = (-2c\partial_\eta)(2c\partial_\xi)u = 0$

which means that $u_{\xi\eta} = 0$ since $c \neq 0$.

The solution of this transformed equation is $u = f(\xi) + g(\eta)$

It agrees exactly with the solution $u(x,t) = f(x+ct) + g(x-ct)$

(3) confirm your answer by direct substitution

$u(x,t) = f(x+ct) + g(x-ct)$.

$u_x = \frac{\partial}{\partial x}[f(x+ct)+g(x-ct)] = \frac{\partial}{\partial x}f(x+ct) + \frac{\partial}{\partial x}g(x-ct) = f'(x+ct)\cdot 1 + g'(x-ct)\cdot 1$

$u_{xx} = \frac{\partial}{\partial x}u_x = \frac{\partial}{\partial x}[f'(x+ct)+g'(x-ct)] = \frac{\partial}{\partial x}f'(x+ct) + \frac{\partial}{\partial x}g'(x-ct) = f''(x+ct)\cdot 1 + g''(x-ct)\cdot 1$

$u_t = \frac{\partial}{\partial t}[f(x+ct)+g(x-ct)] = \frac{\partial}{\partial t}f(x+ct) + \frac{\partial}{\partial t}g(x-ct) = f'(x+ct)\cdot c + g'(x-ct)\cdot(-c)$

$u_{tt} = \frac{\partial}{\partial t}u_t = \frac{\partial}{\partial t}[f'(x+ct)\cdot c - g'(x-ct)\cdot c] = c\cdot\left[\frac{\partial}{\partial t}f'(x+ct) - \frac{\partial}{\partial t}g'(x-ct)\right]$

$= c\cdot[f''(x+ct)\cdot c - g''(x-ct)\cdot(-c)] = c^2[f''(x+ct) + g''(x-ct)]$

$u_{tt} - c^2u_{xx} = c^2[f''(x+ct)+g''(x-ct)] - c^2\cdot[f''(x+ct)+g''(x-ct)] = 0$

Therefore, the solution is $u(x,t) = f(x+ct) + g(x-ct)$

2(c)

From 2(b), $u(x,t) = f(x+ct) + g(x-ct)$

Setting $t=0$, we get $\phi(x) = u(x,0) = f(x) + g(x)$

By the chain rule, $\psi(x) = u_t(x,0) = cf'(x) - cg'(x)$

Hence we have $\begin{cases} \phi' = f' + g' \\ \frac{1}{c}\psi = f' - g' \end{cases} \Rightarrow \begin{cases} f' = \frac{1}{2}(\phi' + \frac{\psi}{c}) \\ g' = \frac{1}{2}(\phi' - \frac{\psi}{c}) \end{cases}$

Integrating, we get $\begin{cases} f(s) = \frac{1}{2}\phi(s) + \frac{1}{2c}\int_0^s \psi + A \\ g(s) = \frac{1}{2}\phi(s) - \frac{1}{2c}\int_0^s \psi + B \end{cases}$ where A and B are constants.

Because of the equation $\phi(x) = f(x) + g(x)$, $A + B = 0$.

So, $u(x,t) = f(x+ct) + g(x-ct) = \frac{1}{2}\phi(x+ct) + \frac{1}{2c}\int_0^{x+ct}\psi + \frac{1}{2}\phi(x-ct) - \frac{1}{2c}\int_0^{x-ct}\psi$

$= \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct}\psi(s)\,ds$

Now, substituting the answer.

① $u(x,0) = \frac{1}{2}[\phi(x) + \phi(x)] + \frac{1}{2c}\int_x^x \psi(s)\,ds = \phi(x)$

② $u_t(x,t) = \frac{1}{2}[\phi'(x+ct)\cdot c + \phi'(x-ct)\cdot(-c)] + \frac{1}{2c}[\psi(x+ct)\cdot c - \psi(x-ct)\cdot(-c)]$

$= \frac{c}{2}[\phi'(x+ct) - \phi'(x-ct)] + \frac{1}{2}[\psi(x+ct) + \psi(x-ct)]$

So, $u_t(x,0) = \frac{c}{2}[\phi'(x) - \phi'(x)] + \frac{1}{2}[\psi(x) + \psi(x)] = \psi(x)$

③ $u_{tt} = \frac{c}{2}[\phi''(x+ct)\cdot c - \phi''(x-ct)\cdot(-c)] + \frac{1}{2}[\psi'(x+ct)\cdot c + \psi'(x-ct)\cdot(-c)]$

$= \frac{c^2}{2}[\phi''(x+ct) + \phi''(x-ct)] + \frac{c}{2}[\psi'(x+ct) - \psi'(x-ct)]$

$u_x = \frac{1}{2}[\phi'(x+ct) + \phi'(x-ct)] + \frac{1}{2c}[\psi(x+ct) - \psi(x-ct)]$

$u_{xx} = \frac{1}{2}[\phi''(x+ct) + \phi''(x-ct)] + \frac{1}{2c}[\psi'(x+ct) - \psi'(x-ct)]$

u_{tt} and u_{xx} satisfy the equation $u_{tt} - c^2 u_{xx} = 0$

2(d) $\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(x,0) = \log(1+x^2), \quad u_t(x,0) = 4+x \end{cases}$

According to 2(c). $u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)\,ds$.

Here, $\phi(x) = \log(1+x^2)$, $\psi(x) = 4+x$.

So, $u(x,t) = \frac{1}{2}\{\log[1+(x+ct)^2] + \log[1+(x-ct)^2]\} + \frac{1}{2c}\int_{x-ct}^{x+ct} (4+s)\,ds$

$= \frac{1}{2}\log[1+(x+ct)^2][1+(x-ct)^2] + \frac{1}{2c}(4s + \frac{1}{2}s^2)\Big|_{x-ct}^{x+ct}$

$= \log\sqrt{[1+(x+ct)^2][1+(x-ct)^2]} + t(4+x)$.